# Bank Loan Performance Analysis Report (Power BI)

## Executive Summary

This report delivers a comprehensive, interactive analysis of the bank's loan portfolio, covering a total of 38,576 loan applications and $435.8 Million in funded amount. The primary goal is to provide a single source of truth for tracking key lending metrics, identifying high-risk borrower segments, and supporting strategic decision-making in credit risk management.

**Key Insight:** The overall Bad Loan Rate stands at 13.82%, with Debt Consolidation and 60-month terms identified as the highest-risk segments requiring immediate policy review.

## Section 1: Data & Technical Methodology

### 1.1 Data Source and Volume

- **Source:** Simulated Financial Loan Dataset (financial_loan.csv).
- **Data Volume:** 38,576 individual loan records.
- **Time Period:** Data spans primarily across 2021.

### 1.2 Data Preparation (SQL/Python & Power Query)

The raw data was transformed to ensure analytical readiness, including:

- **Risk Categorization:** Created a new custom field, **Loan_Status_Category**, to classify original statuses (e.g., 'Charged Off', 'Default') into a simplified **"Bad Loan"** vs. **"Good Loan"** for high-level KPI tracking.
- **Date Formatting:** Consolidated issue_date into a clean Issue_Month_Year column for trend analysis.
- **Aggregation:** Pre-aggregated segmentation tables (e.g., purpose_segment.csv, monthly_trend.csv) were created and loaded to optimize dashboard performance.

## Section 2: Summary Dashboard Analysis (Portfolio Health)

The Summary Dashboard serves as the first point of contact for monitoring core performance metrics and portfolio risk.

### 2.1 Overall Performance Metrics

The dashboard highlights strong repayment performance with a positive return:

- **Total Funded Amount: $435.8 Million**
- **Total Amount Received: $473.1 Million**
- **Average Interest Rate: 12.05%**

### 2.2 Credit Risk Exposure

The primary concern identified in the Summary view is the Bad Loan ratio:

- **Bad Loan Count: 5,333**
- **Bad Loan Percentage: 13.82%** (5,333 / 38,576)

### 2.3 Insights from Loan Status Breakdown

The interactive **Loan Status Table** reveals the core problem: The highest count of bad loans is categorized under **'Charged Off'** (loans deemed uncollectible), emphasizing the need for stricter underwriting prior to this severe stage. The Slicers (Grade and Term) allow instant deep-dives into which loan grades or terms drive this 13.82% rate.

## Section 3: Overview Dashboard Analysis (Segmentation)

The Overview Dashboard breaks down performance by borrower and loan characteristics to pinpoint specific risk drivers.

### 3.1 Monthly Funding Trend

The Line Chart shows a clear **acceleration in loan volume** in the latter months of the year, with **November 2021** being the peak month. This trend necessitates that risk management teams ensure the quality of loans kept pace with the rapid volume increase.

### 3.2 High-Risk Segment Analysis

| **Segment** | **Total Funded Amount** | **Average Interest Rate** | **Key Insight** |
|---|---|---|---|
| **Loan Purpose** | Debt Consolidation: **$232.5M** | **12.50%** (Highest Rate) | Represents over half of total funding and carries the highest average interest rate, making it the **highest-risk category** by volume and cost. |
| **Home Ownership** | Rent: **$185.8M** | **12.30%** (Higher than Mortgage) | Renters account for a large portion of the funding and are assigned a notably higher average interest rate than Mortgaged borrowers (**11.80%**), indicating a clear risk assessment by lenders. |
| **Loan Term** | 60 Months: **$162.7M** | **14.83%** (Significantly Higher) | Loans with the longer 60-month term carry an average interest rate **~3.8 percentage points higher** than 36-month loans (11.03%), directly correlating longer term with higher inherent risk. |

### 3.3 Geographic Concentration

The **Geographic Map** demonstrates that lending activity is highly concentrated in large states, particularly **California** and **Texas**, suggesting that risk profiles and marketing strategies should be optimized for these specific regions.

## Section 4: Conclusion and Recommendations

The completed Power BI solution provides a robust platform for data-driven credit risk management. The analysis confirms strong overall volume but highlights concentrated risk in specific segments.

**Actionable Recommendations:**

1. **Review Debt Consolidation Policy:** Given that Debt Consolidation is the largest funded segment ($232.5M) and carries the highest average interest rate (12.50%), conduct a targeted study on the default rate specifically within this purpose category to re-evaluate underwriting criteria.
2. **Monitor 60-Month Term Loans:** Implement a higher scrutiny and faster review process for 60-month term applications due to their significantly higher average interest rate (**14.83%**), which flags them as the riskier lending product.
3. **Optimize Regional Risk Models:** Leverage the **address_state** segmentation to create region-specific risk bands, ensuring that the high concentration of loans in key states (like CA and TX) is managed with localized insights.
4. **Utilize Details Dashboard for Operational Audits:** Empower the collections and loan servicing teams to use the **Details Dashboard** for targeted lookups and data exports when managing individual delinquent accounts, streamlining the operational process.